MUHAMMAD SOHAIL

Machine Learning Engineer | Data Science Graduate

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PROFESSIONAL SUMMARY

Entry-level Machine Learning Engineer candidate (BS Data Science, GCUF 2025) with hands-on project experience building and evaluating machine learning models in Python using Pandas, Scikit-learn, and TensorFlow/Keras. Projects include an NLP resume classifier, a customer churn model with class imbalance handling (SMOTE), and a neural network for price prediction, each evaluated with standard metrics and delivered as a working app. Internship experience in data science, AI/ML, and Generative AI. Seeking to ship practical, well-evaluated ML features.

TECHNICAL SKILLS

Programming: Python, SQL
Machine Learning: Scikit-learn, XGBoost, CatBoost, Random Forest, Feature Engineering, Class Imbalance Handling (SMOTE), Exploratory Data Analysis (EDA), Model Evaluation (Precision, Recall, F1-Score, ROC-AUC, MAE, MSE, R-squared)
Deep Learning: TensorFlow/Keras (Artificial Neural Networks), Early Stopping
NLP & Generative AI: NLTK, TF-IDF, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, ChromaDB, FAISS, sentence-transformers
Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI
Backend & Deployment: FastAPI, Flask, REST APIs, Streamlit
Tools: Git, GitHub
Soft Skills: Problem-Solving, Quick Learner, Attention to Detail

EXPERIENCE

Data Science Intern | 10Pearls | 11/2026 – 12/2026
- Working on assigned data science tasks in Python, covering data preprocessing, exploratory analysis, and modeling.
- Applying supervised machine learning and model evaluation techniques to assigned data science problems.

AI/ML Intern | Oasis Infobyte | 09/2026 – 11/2026
- Completing assigned AI/ML and Data Science projects in Python covering data preprocessing, model development, and evaluation.

AI Intern | CodeAlpha | 08/2026 – 09/2026
- Selected for the Machine Learning and Generative AI Internship Program; worked on applied ML and Generative AI projects under mentorship.

PROJECTS

AI Resume Scanner | github.com/sohail189/resume-scanner-system
Python, NLTK, TF-IDF, Scikit-learn, PyPDF2, python-docx, Streamlit
- Developed an NLP pipeline that extracts text from PDF and DOCX resumes, preprocesses it with NLTK, and vectorizes it with TF-IDF.
- Classified resumes into job categories using a Scikit-learn classifier; delivered as a Streamlit app for resume upload and analysis.

Customer Churn Prediction | github.com/sohail189/churn-prediction
Python, Pandas, Scikit-learn, XGBoost, SMOTE, Streamlit
- Built a churn classification pipeline with missing-value handling, categorical encoding, and SMOTE for class imbalance.
- Trained and compared Logistic Regression, Random Forest, and XGBoost using precision, recall, F1-score, and ROC-AUC; delivered as a Streamlit app.

House Price Prediction | github.com/sohail189/house-price-prediction
Python, TensorFlow/Keras, Flask, Streamlit
- Built an Artificial Neural Network (RMSprop, early stopping) to predict California house prices, evaluated with MAE, MSE, MSLE, and R-squared; delivered through Flask and Streamlit interfaces.

Heart & Liver Disease Prediction | github.com/sohail189/Webapp.py
Python, CatBoost, Random Forest, Streamlit
- Developed CatBoost and Random Forest classifiers for heart and liver disease risk, with an interactive Streamlit EDA dashboard.

Pakistan Tourist RAG System | github.com/sohail189/Pakistan-Tourist-RAG-System
Python, FastAPI, Streamlit, LangChain, ChromaDB, FAISS, sentence-transformers, Groq API, Gemini API
- Built a Retrieval-Augmented Generation (RAG) application that answers tourism queries about Pakistan using sentence-transformers embeddings, ChromaDB retrieval, and Llama 3.3 70B (Groq API); implemented a second variant with FAISS and the Gemini API over 110 knowledge chunks.
- Exposed the pipeline through a FastAPI REST backend and a Streamlit interface.

EDUCATION

Bachelor's in Data Science (BS) | Government College University Faisalabad (GCUF) | 2021 – 2025